

Evaporation Cooling System

UNR CPE 301 Final Project

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Github Link: [UNR CPE 301 Final](#)

Project Description

This project functions as the embedded system for an evaporation cooling system (swamp cooler) through the use of an arduino and various components. The project has the ability to toggle a fan depending on the monitored temperature as well as enable or disable through buttons. It displays the current temperature and humidity on an LCD screen while enabled and an error message when the monitored water level is too low. The project also has the capability of recording a timestamp to the computer for each state change.

States

The project includes four unique states, where in all but the disabled state the humidity and temperature are monitored and displayed on the screen, user input from the stop button transitions to the disabled state, and the changes in the vent position from the input of the potentiometer are recorded to the computer. The RTC is used to timestamp these changes as well as the transitions to other states.

Disabled

This is the initial state of the project and can be reached by the user pressing the stop button. The yellow LED will be lit, temperature and water level will not be monitored, and input from the start button is monitored using an interrupt service routine.

Idle

The idle state is reached when the user presses the start button. In this state, the water level is monitored to where the state will change to error if it dips below the threshold. The green LED is lit to indicate this state. If the monitored temperature rises above the threshold, the project will transition from this state to running.

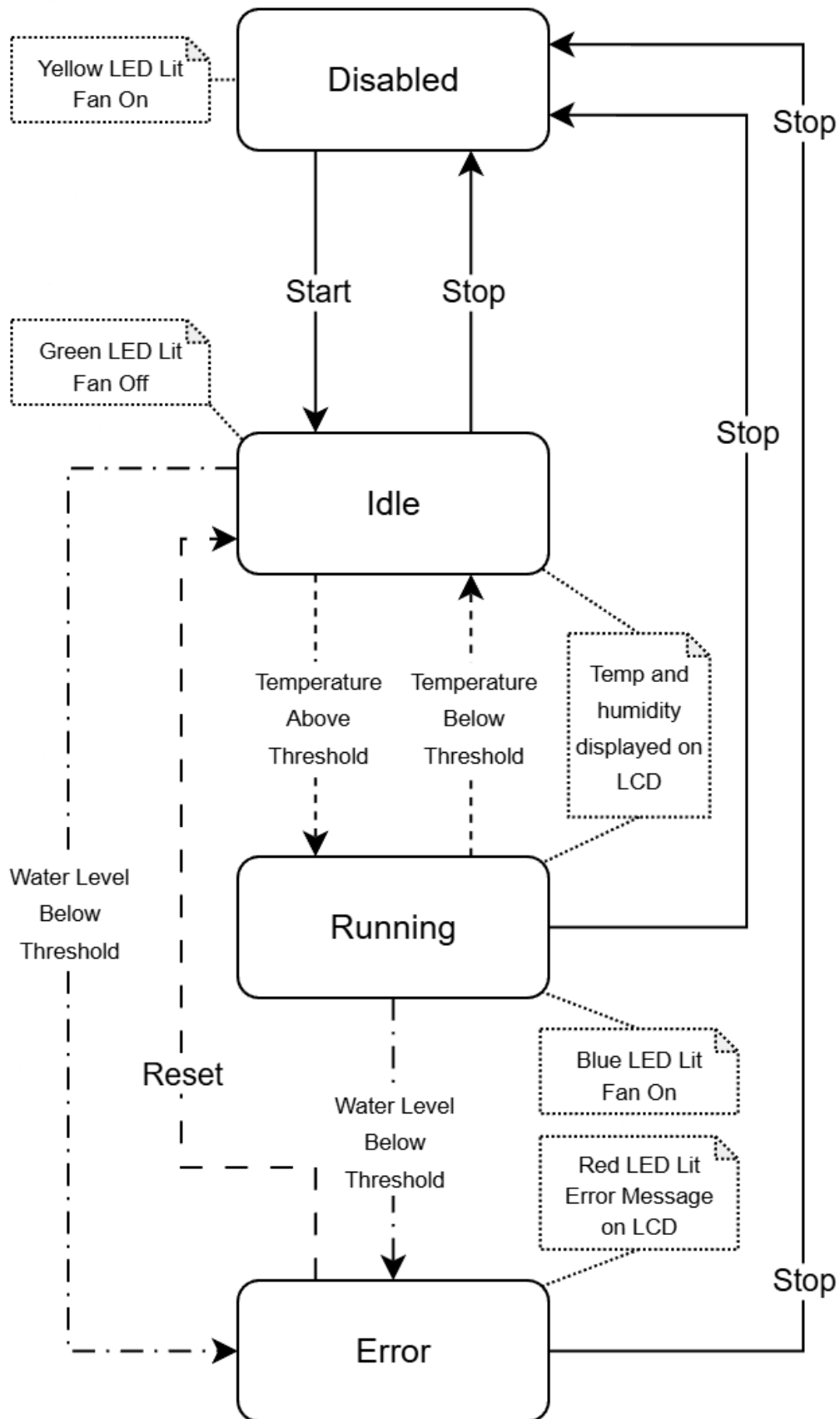
Error

The error state is reached whenever the water level sensor detects that the water level is too low. The red LED is lit and an error message is displayed to the LCD. To leave this state when the reset button is pressed, if the water level has risen back above the threshold, it will transition to the idle state.

Running

In the running state, the fan is toggled on along with the blue LED. If the monitored temperature drops below the threshold the system transitions to the idle state, and if the water level drops too low, it transitions to the error state.

State Diagram



Components

Arduino Mega 2560

This is the computer of the system which interfaces with all of the other components of the project and applies the logic to make the project function.

Breadboard Power Supply Module

The breadboard power supply module provides external power to the breadboards to prevent damage to the output circuitry of the Arduino. This is required to power both the stepper motor and the DC motor.

DC Motor / L293D Motor Controller

Using a DC motor in this project allows for the L293D Motor Controller to be implemented, allowing a fan to be easily operated by the Arduino. This fan is used as a model to move the cool air from the system when turned on in the running state.

DHT11 Temperature and Humidity Sensor

This sensor allows the recording of the current temperature and humidity of its environment which is used to determine when the system should be enabled to cool the room down.

LCD Display

The 16 pin LCD Display allows the current state to be displayed in the case of idle and running or for the current state to be displayed in the case of error or disabled.

LEDs

Yellow, green, red, and blue LEDs are used as a visual representation for the current state. The states of disabled, idle, error, and running correlate to the yellow, green, red, and blue LEDs being on.

Power Supply Module

Running both motors along with all of the other components could cause damage to the power supply circuitry in the Arduino, so the MB102 Breadboard Power Supply Module is used to supplement the extra power to the motors.

Push Buttons

The push buttons get input from the user to go between the states. The start button applies an interrupt to enable the system from disabled, with the stop button disabling the system from idle or running. When pressed, the reset button goes back to the idle state from error if the water level has been corrected.

Real-Time Clock Module

The DS1307 RTC is used for the logs to accurately report the timestamps for each state transition as well as for the changes in vent position.

Resistors

The project contains resistors to limit the voltage going to the LEDs to prevent them from breaking as well as pull-down resistors for the buttons to prevent floating voltage.

28BYJ-48 Stepper Motor / 28BYJ-48 Motor Driver

The direction of the output vent is controlled by the stepper motor, which is adjustable in all states but disabled. The motor driver is used to control the stepper motor by interfacing it with the Arduino.

Trimmer Potentiometer

In order for the vent position to be changed, the trimmer potentiometer is mapped to the position of the stepper motor so that when the user turns the potentiometer, the stepper motor's position changes. Another trimmer potentiometer is also used for the LCD to adjust its contrast.

Water Level Sensor

This sensor monitors the level of water in a reservoir so that when the water level drops below the threshold, the system goes to the error state. This makes it so that the fan is not running without any evaporation occurring in the system.

System Overview

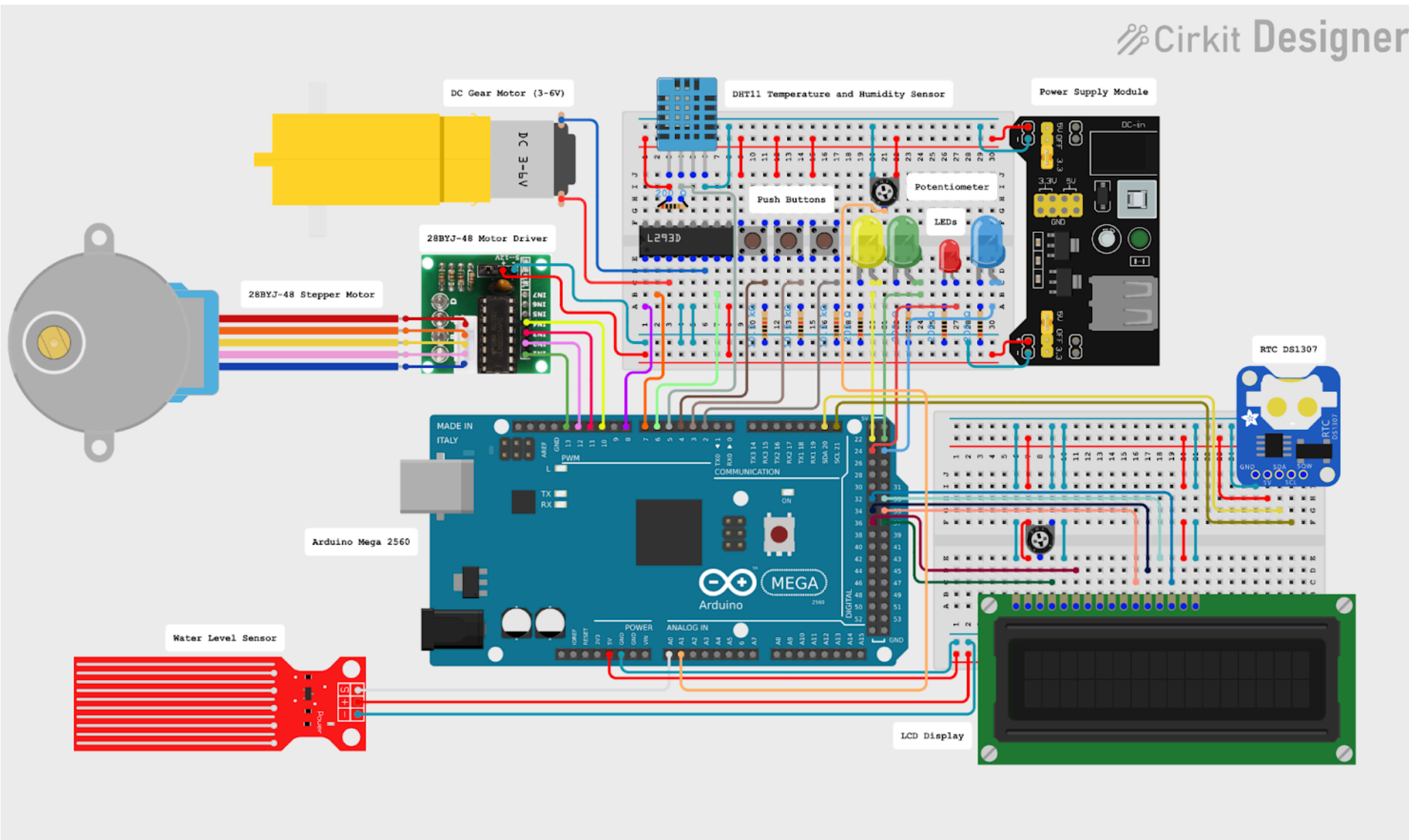
The system requires a wall outlet to power the external breadboard power supply module as well as a USB connection to a computer for power and serial monitor output. The system begins running at 25°C and requires some water in the reservoir to run.

Specification Sheets

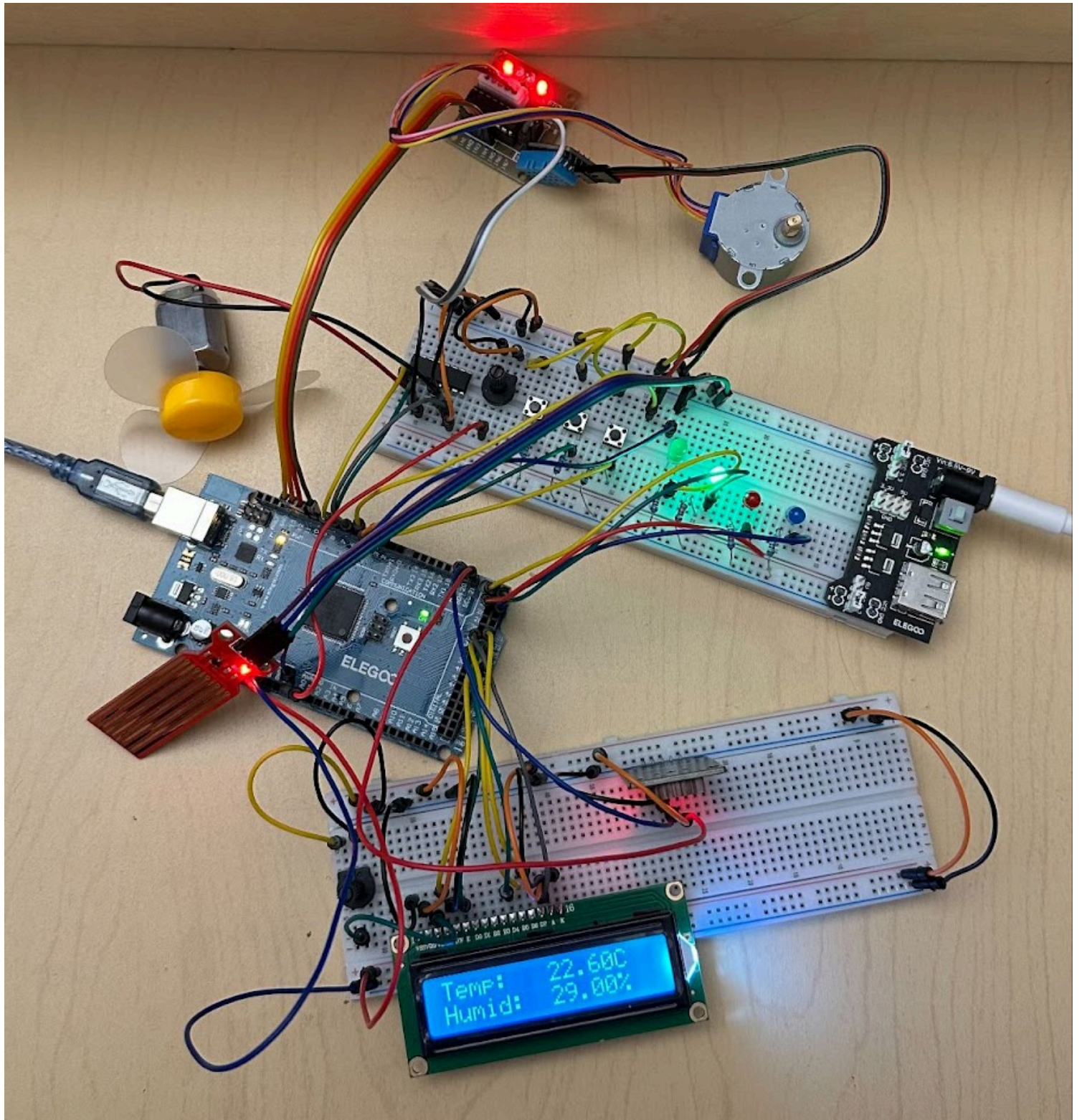
The specifications sheets for all components can be found in [this linked folder](#).

Schematic Diagram

Circuit Designer



Circuit Image



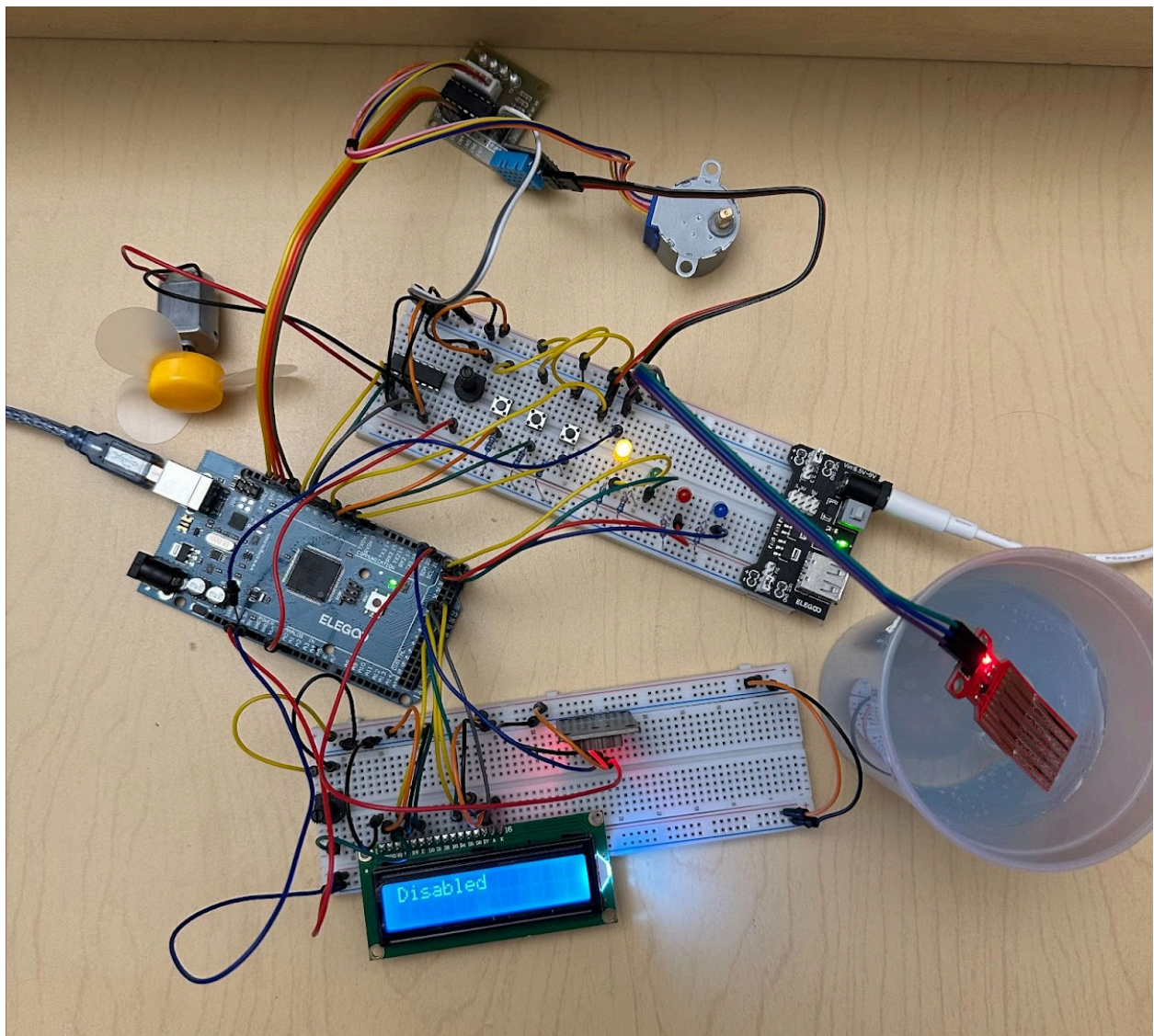
System Demonstration

Operation Video

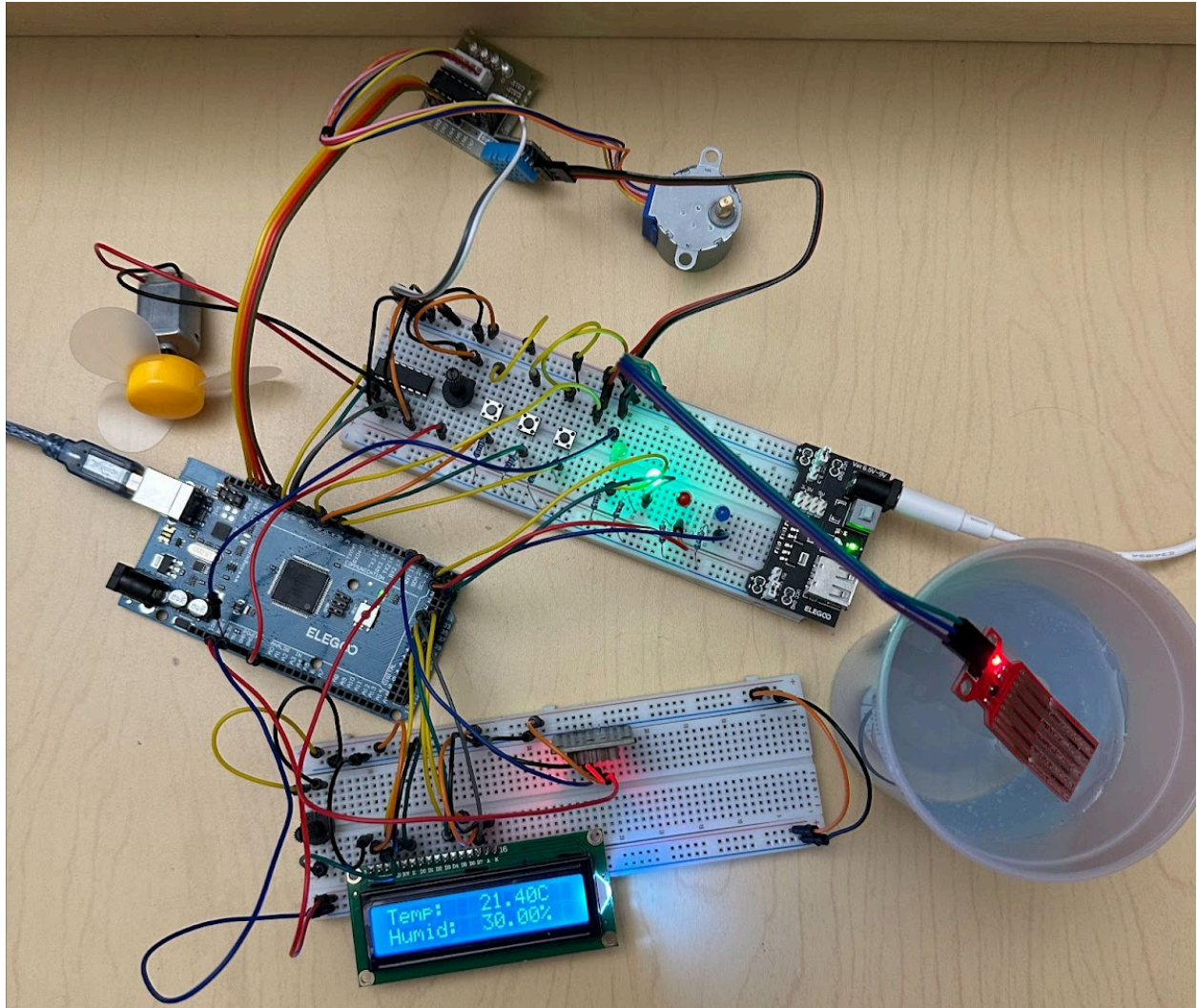
This is a link to the video of the project: <https://youtu.be/PBWLqjcgNgw>

State Images

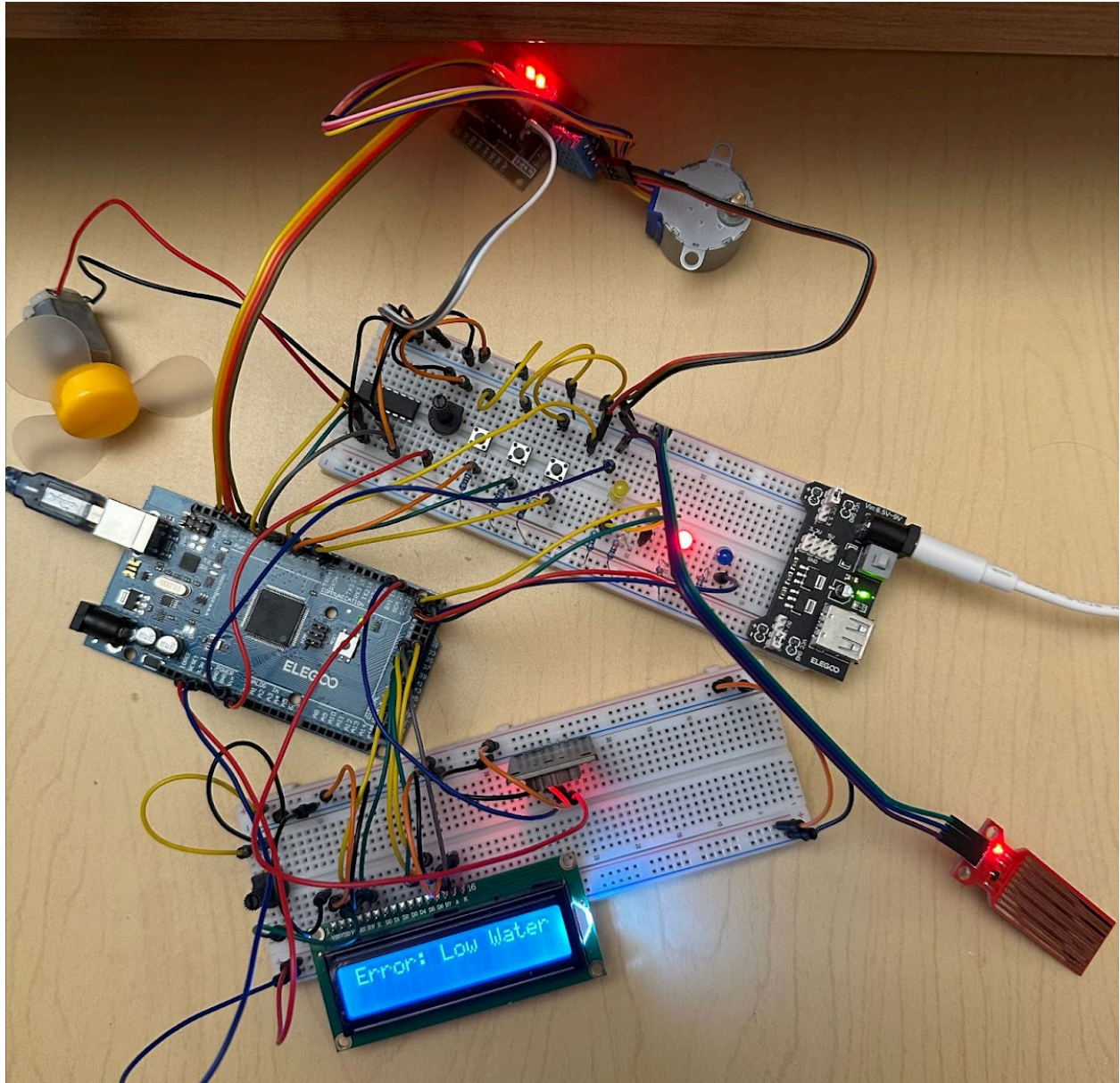
Disabled



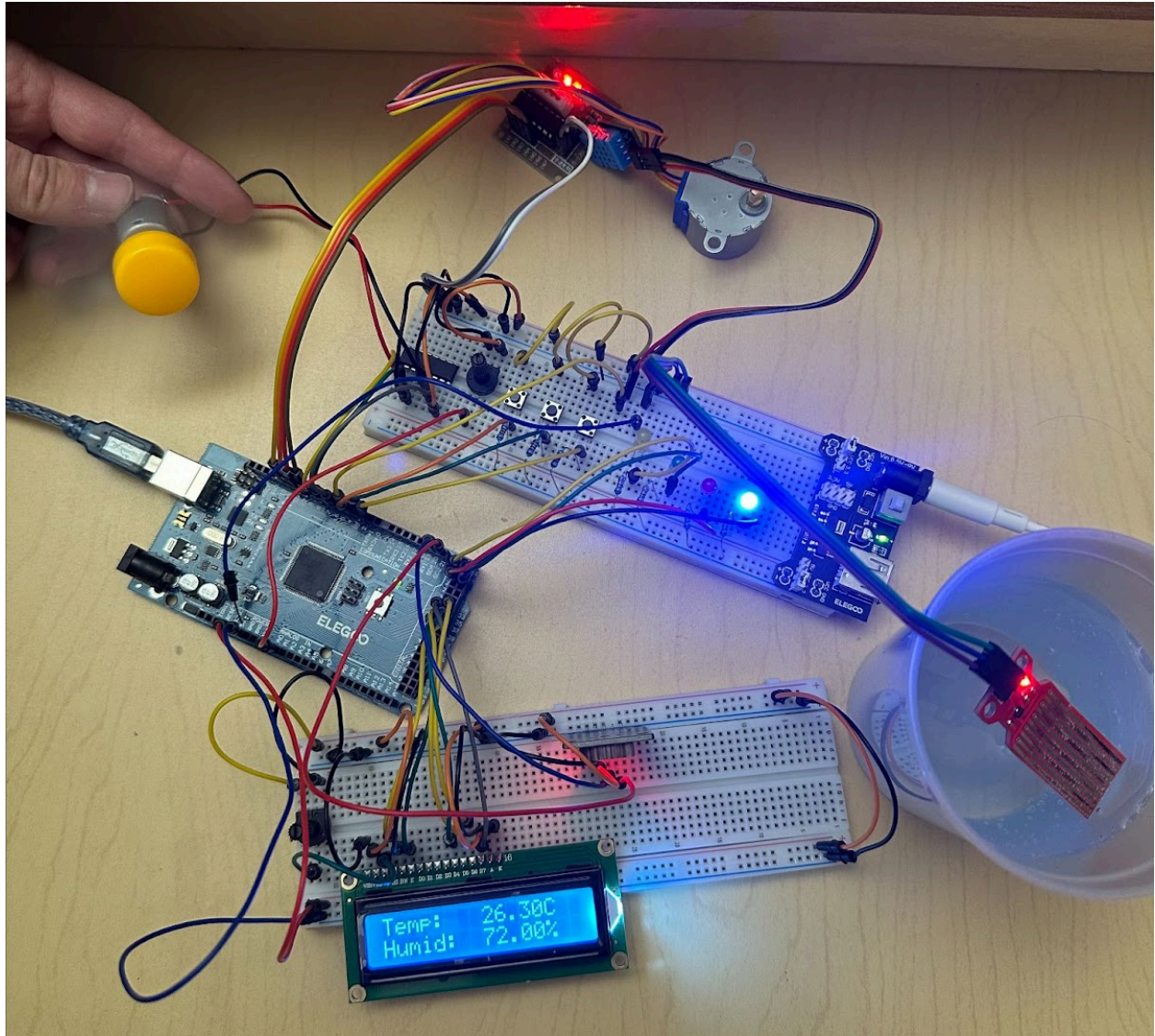
Idle



Error



Running



Environmental Impact

This system was created with environmental impact and sustainability at the heart of the design philosophy.

One example of this can be seen in the energy efficiency of the system. For example, an interrupt service request was used to monitor the start button. This is far more power efficient than using polling to continually check the state of the button, as this would result in an extra check in every cycle of the microcontroller's operation. Another example of this power efficiency can be seen in how the various components are only read and utilized within the states they are needed. For example, both the vent fan and water sensor only have their values measured when the system is not in the disabled state, saving more instruction calls. Another aspect this system incorporates is design safety. The system utilizes low voltage (5V) components, as well as current limiting resistors in places where they are necessary, such as in series with the four LEDs to prevent burnout. Further, the system utilizes an error state that prevents all component functionality when an unintended state is reached. The system will continue to monitor the components such as the water and heat sensors, and a reset button will return it back to a functioning state if and only if these values are back within an intended range. This prevents any unintended consequences that may result from such states.

Affordability, Sustainability, and Accessibility

The materials and design decisions of the system were chosen to promote the principles of affordability, sustainability, and accessibility. The kit including the components as well as the microcontroller cost just under \$70, and included more components than were used in building the system, meaning that the total cost was even less than this. The system used inexpensive, modular, low voltage components, which contributed both to its affordability and ease of assembly. The highly modular design of the system also means that if one component breaks, only that component needs to be replaced rather than the entire system. This means that it is cheap to repair and doesn't create a large amount of material waste, as the system can be in commission for an extended period of time with only small component replacements. Further, the materials used were, although inexpensive, still relatively durable, meaning that repairs would not need to happen often. Lastly, the user interface of the system is exceedingly simple, consisting only of three buttons to start, stop, and reset the system, as well as four LEDs to describe the current state. It also uses an LCD screen to describe the current humidity and temperature of the system, and logs state and motor transitions for the user.